FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – III
PAPER –2
TEST DATE: 18-01-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

Section – A ***(01 –06, 20 – 25, 39 – 44)*****:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A ***(07 –10, 26 – 29, 45 – 48)*****:** This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Section – B ***(11 – 13, 30 – 32, 49 – 51)*****:** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

Section – C ***(14 – 19, 33 – 38, 52 – 57)*****:** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

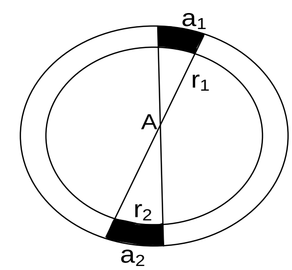
Physics

PART – I

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A wire having a uniform linear charge density λ, is bent in the form of a ring of radius R. Point A as shown in the figure, is in the plane of the ring but not at, the centre. Two elements of the ring of lengths a_1 and a_2 subtend very small and equal angles at the Point A. They are at distances r_1 and r_2 from the point A respectively.



(A) The ratio of charge on elements a_1 and a_2 is $\frac{r_1}{r_2}$.

(B) The element a_1 produces greater magnitude of electric field at A than element a_2.

(C) The element a_1 and a_2 produce same potential at A.

(D) The elements a_1 produces greater potential at A than element a_2.

Ans. ABC

Sol. $dq_1 = \lambda a_1$

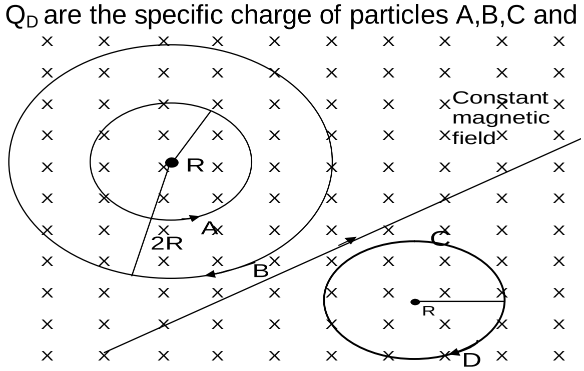
$dq_2 = \lambda a_2$

$$\frac{dq_1}{dq_2} = \frac{a_1}{a_2} = \frac{\theta\, r_1}{\theta\, r_2} = \frac{r_1}{r_2}$$

$$E_1 = \frac{kdq_1}{r_1^2} = \frac{k\lambda dr_1}{r_1^2} = \frac{\lambda\theta}{r_1}$$

$$E_1 \propto \frac{1}{r_1}$$

2. Four particles A, B, C and D of masses m_A, m_B, m_C and m_D respectively, follow the paths shown in the figure, in a uniform magnetic field. Each particle moving with same speed. Q_A, Q_B, Q_C and Q_D are the specific charge of particles A,B,C and D respectively :



(A) $Q_A < Q_B < Q_C < Q_D$

(B) $Q_D < Q_B < Q_C < Q_A$

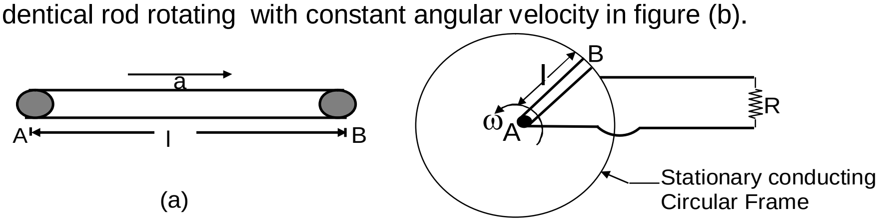
(C) Charge on the particle B and particle D is of different nature

(D) Work done by magnetic force on the particle C is minimum as compared to other particle

Ans. B

Sol. Since B & revolve in opposite since, thus they are oppositely charged. From direction of motion of the charges Q_A = + ve Q_B & $Q_D \rightarrow$ -ve $Q_C \rightarrow$ neutral

3. consider a metal rod of length L that is given a uniform acceleration as shown in figure (a) and an identical rod rotating with constant angular velocity in figure (b).



(a) (b)

(A) If V_A and V_B are potentials of end A and B respectively, then $V_A < V_B$ in figure (a)

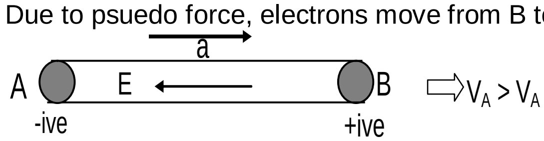
(B) If V_A and V_B are potentials of end A and B respectively, then $V_A > V_B$ in figure (b)

(C) Electric field inside rod has magnitude $\frac{ma}{e}$ in figure (a)

(D) Electric field inside rod has magnitude $\frac{m\omega^2 r}{e}$ in figure (b)

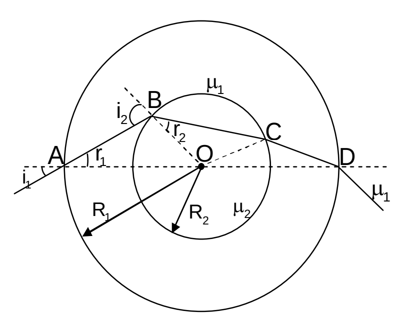
Ans. ABCD

Sol. Due to psuedo force, electrons move from B to A.



In figure (b) electron moves from A to B
Region I Region II

4. There is a spherical glass ball of refractive index μ_1 and another glass ball of refractive index μ_2 inside it as shown in figure. The radius of the outer ball is R_1 and that of inner ball is R_2. A ray is incident on the outer surface of the ball at an angle i_1. Mark the correct options.



(A) The Value or r_1 is $\sin^{-1}\left(\frac{\sin i_1}{\mu_1}\right)$

(B) The value of r_2 is $\sin^{-1}\left(\frac{R_1}{\mu_2 R_2}\sin i_1\right)$

(C) The value of r_1 is $\sin^{-1}\left(\frac{\sin i_2}{\mu_2}\right)$

(D) The value of r_2 is $\sin^{-1}\left(\frac{R_2}{\mu_2 R_1}\sin i_1\right)$

Ans. AB

Sol. $\mu_1 \sin r_1 = \sin i_1$

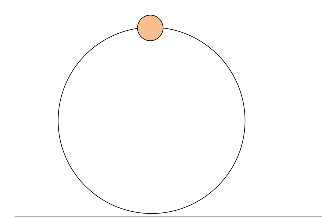
$\Rightarrow \quad r_1 = \sin^{-1}\left(\frac{\sin i_1}{\mu_1}\right)$

$\frac{\mu_1}{\mu_2}\sin i_2 = \sin r_2$

$\frac{\mu_1}{\mu_2}\frac{R_1}{R_2}\frac{\sin i_1}{\mu_1} = \sin r_2$

$r_2 = \sin^{-1}\left(\frac{R_1}{\mu_2 R_2}\sin i_1\right)$

5. A bead is fixed on a rigid ring of radius r and of mass that negligible as compared to that of the bead. Assembly thus formed is held motionless on a horizontal floor with the plane of the ring vertical and the bead on the top. Coefficient of friction between the ring and the floor is μ. Now the assembly is given a slight push so that the ring starts rolling Acceleration of free fall is g. Choose the correct statements.



(A) The angle rotated by the ring before it begins to slide $\tan^{-1}(\mu)$

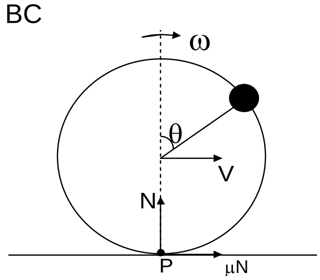
(B) The speed of centre of the ring before it begins to slide $\mu\sqrt{gR}$

(C) The angle rotated by the ring before it begins to slide $2\tan^{-1}(\mu)$

(D) The speed of the centre of the ring before it begins to slide. $\mu\sqrt{2gR}$

Ans. BC

Sol.



V_C is Horizontal only.

$\overrightarrow{V_P} = \overrightarrow{V_{PC}} + \overrightarrow{V_C}$

$V_P = 0$

Mechanical energy conservation,

$mgR(1-\cos\theta) = \frac{1}{2}mV_{bead}^2 \qquad V_{Bead} = 2v\cos\theta/2$

$gR \times 2\sin^2\theta/2 = 2v^2\cos^2\theta/2$

$V = \sqrt{gR}\tan\theta/2$

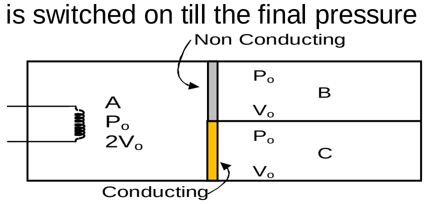
$\overrightarrow{\tau_{em}} = I_{cm} \propto = 0 \quad$ As $I_{cm} = 0$

$NR\sin\theta = \mu N(R + R\cos\theta)$

$\tan\theta/2 = \mu$

$\theta = 2\tan^{-1}(\mu)$

6. An adiabatic container of volume 4 V_0 is divided into two equal parts by rigid fixed wall whose lower half is conducting and upper half in non-conducting. Initially the right portion is divided into two equal parts by a freely moving massless & non conducting piston. Section A has 2 moles, B and C each has one mole of an ideal gas with adiabatic exponent $(\gamma)1.5$. If the heater in left part is switched on till the final pressure C becomes (125/27) Po. Mark the correct statements



(A) Final temperature in section A $\frac{205P_oV_o}{27R}$

(B) Final temperature in section B $\frac{5P_oV_o}{3R}$

(C) Final temperature in section C $\frac{205P_oV_o}{27R}$

(D) Heat supplied by the Heater $\frac{368}{9}P_oV_o$

Ans. ABCD

Sol. For piston to be in equilibrium

$P_B = P_C$

Process in B is adiabatic,

Process in A is Isochoric

At every instant Temperature of A = Temperature of C

For B

$P_oV_o^{1.5} = \frac{125}{27}P_ov^{1.5} \qquad V = \frac{9V_o}{25}$

Volume of C will increase by $\frac{16V_o}{25}$

$V_C = \frac{41V_o}{25}$

$T_B = \frac{125}{27}P_o\frac{9V_o}{25} = \frac{5P_oV_o}{3R}$

$T_C = \frac{205}{27}\frac{P_oV_o}{R} - = T_A$

Net work done by gas is zero as there is no pushing of surroundings.

Q Heater = ΔU_{syst}

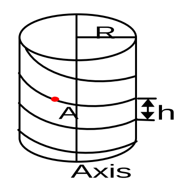
Q Heater = $\frac{368}{9}P_oV_o$

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 07 and 08

A small bead starts sliding down a frictionless rigid helical wire frame. The axis of the helix is vertical, radius is R and pitch is h.



7. In How much time, will the Bead descend a Height H?

(A) $\sqrt{\frac{2H}{g}}\sqrt{\frac{h^2+4\pi^2R^2}{h}}$

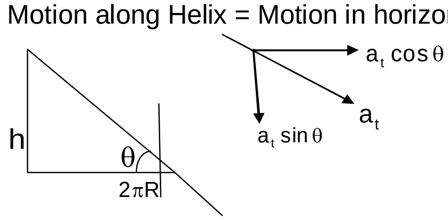
(B) $\sqrt{\frac{H}{g}}\frac{\sqrt{h^2+4\pi^2R^2}}{h}$

(C) $\sqrt{\frac{H}{g}}\frac{\sqrt{h^2+4\pi^2R^2}}{2h}$

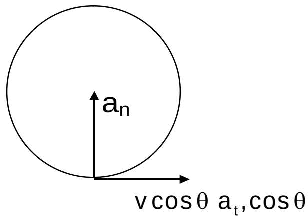
(D) None of these

Ans. A

Sol. Motion along Helix = Motion in horizontal plane + Motion along Axis.



Top view



$a_t = g\sin\theta$

A along Axis = $a_t \sin\theta = g\sin^2\theta$.

$H = \frac{1}{2} g\sin^2\theta$. t = $\sqrt{\frac{2H}{g}} \times \frac{1}{\sin\theta} = \sqrt{\frac{2H}{g}} \frac{\sqrt{h^2 + 4\pi^2 R^2}}{h}$

8. How much force will the wire frame exert on the bead when the bead has descended a height H? [Acceleration of free ball g]

(A) $2\pi mgR \frac{\sqrt{h^2 + 4\pi^2 R^2 + 16\pi^2 H^2}}{h^2 + 4\pi^2 R^2}$

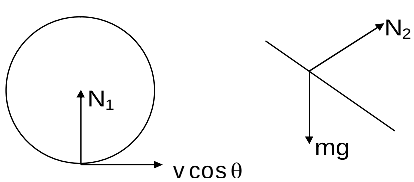
(B) $\frac{\pi mgR\sqrt{h^2 + 4\pi^2 R^2 + 16\pi^2 H^2}}{h^2 + 4\pi^2 R^2}$

(C) $\frac{\pi mgR\sqrt{h^2 + \pi^2 R^2 + 4\pi^2 H^2}}{h^2 + 4\pi^2 R^2}$

(D) None of these

Ans. A

Sol.



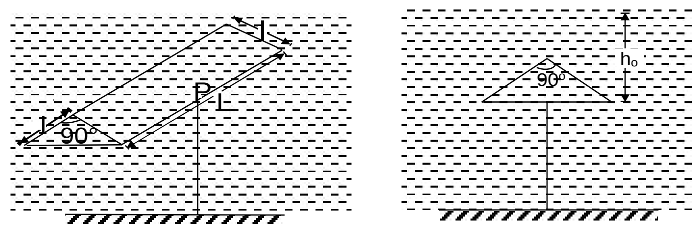
N = $\sqrt{N_1^2 + N_2^2}$ $N_1 = \frac{m(v\cos\theta)^2}{R}$ $N_2 = mg\cos\theta$

$v = \sqrt{2gH}$

N = $\frac{2\pi mgR\sqrt{h^2 + 4\pi^2 R^2 + 16\pi^2 H^2}}{h^2 + 4\pi^2 R^2}$

Paragraph for Question Nos. 09 and 10

A prism shaped Styrofoam of density $\rho_{styrofoam} < \rho_{water}$ is held completely submerged in water. It lies with its base horizontal. The base of foam is at a depth h_0 below water surface and atmospheric pressure is P_o. Surface is open to atmosphere. Styrofoam prism is held in equilibrium by the string attached symmetrically.
$\left(\text{Take}: \rho_{styrofoam} = \rho_f;\ \rho_{water} = \rho_\omega\right)$.



9. Net force exerted by liquid on the Styrofoam is:

(A) $\sqrt{2}\rho_\omega g l^2 L$

(B) $2\rho_\omega g l^2 L$

(C) $\rho_\omega g \frac{l^2 L}{2}$

(D) $\rho_\omega g l^2 L$

Ans. C

10. Magnitude of force on any one of the slant face of Styrofoam is:

(A) $\left[P_o + \rho_\omega g\left(h_o - \sqrt{2l}\right)\right]Ll$

(B) $\left[P_o + \rho_\omega g\left(h_o - \frac{l}{\sqrt{2}}\right)\right]Ll$

(C) $\left[P_o - \rho_\omega g\left(h_o - \frac{l}{\sqrt{2}}\right)\right]Ll$

(D) $\left[P_o + p_\omega g\left(h_o - \frac{l}{2\sqrt{2}}\right)\right]Ll$

Ans. D

Sol. Net force exerted by liquid on Styrofoam is buoyant force = $\rho_\omega g \frac{l^2 L}{2}$

Average pressure on slant surface

$$P_{avg} = \frac{\left(P_o + \rho_\omega g h_o\right) + \left(P_o + \rho_\omega g h_o - \frac{l}{\sqrt{2}}\right)}{2}$$

$$= \left(P_o + \rho_\omega g h_o - \frac{1}{2\sqrt{2}} \right)$$

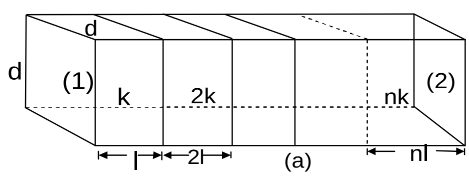
Force on any one of the slant surface

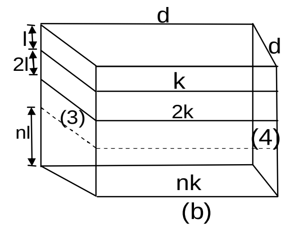
$$\left[P_o + \rho_\omega g \left(h_o - \frac{l}{2\sqrt{2}} \right) \right] Ll$$

Section – B (Maximum Marks: 12)

This section contains ***THREE (03)*** *questions. The answer to each question is a* ***NON-NEGATIVE INTEGER.***

11. An arrangement consisting of slabs is show in the figure. When system is placed in situation (a) with temperature difference is maintained between (1) and (2), equivalent thermal conductivity of system is K and when system in changed to situation (b) and temperature difference is maintained between (3) and (4) then equivalent thermal conductivity increases by 20%. Find number of slabs in the arrangement.





Ans. 4

Sol. In first case

$$\frac{l + 2l + ... + nl}{k_{eq}A} = \frac{l}{kA} + \frac{2l}{3kA} + \frac{3l}{3kA} + ... + \frac{nl}{nkA}$$

$$k_{eq_1} = \frac{(n+1)k}{2}$$

In second case

$$k_{eq_2} = \frac{\sum k_1 Ai}{\sum A_i}$$

$$= \frac{kA' + (2k)(2A') + ... + (nk)(nA')}{A' + 2A' + ... + nA'}$$

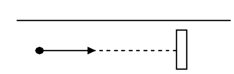
$$= \frac{(2n+1)k}{2}$$

$$\frac{(2n+1)k}{3} = \frac{6}{5}\frac{(n+1)k}{2}$$

9n + 9 = 10n + 5

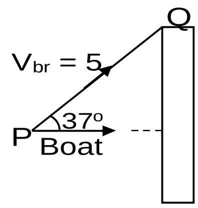
n = 4

12. A small boat is travelling downstream with velocity 5 m/s relative to river flowing with speed 7/3 m/s. it encounters a log floating in the river at some distance from itself. It turns by an angle of 37^o and just crosses the edge of the log after 8 sec. Find time taken by boat to just cross the edge of log if log is kept fixed.



Ans. 6

Sol.



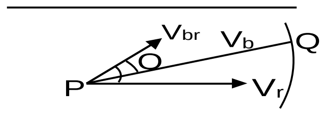
PQ = $5 \times 8 = 40$m
w.r.t Ground

$\overrightarrow{V_b} = \overrightarrow{V_{br}} + \overrightarrow{V_r}$

Velocity must be along PQ

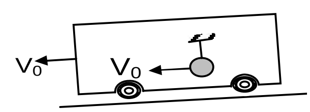
$v_{br} \sin\theta = v_r \sin 37$

$\sin\theta = 7/25$



$$t = \frac{40}{v_{br}\cos\theta + v_r \cos 37^\circ} = 6 \text{ sec.}$$

13. A pendulum with massless string of length l and bob of mass m is attached inside a railway carriage which is moving with speed v_0 as shown. Carriage is brought to rest uniformly by applying brakes. When carriage comes to rest pendulum string is horizontal. Minimum magnitude of retardation of carriage such that bob completes vertical circular motion is (3k-2) ms^{-2}. Find k. (g = 10 ms^{-2})



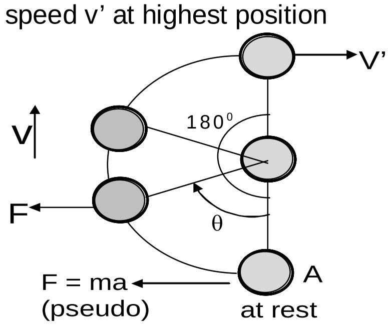
Ans. 9

Sol. Acceleration of train = $a \longrightarrow$

Motion of bob w.r.t. train:

Initially $\vec{V}_{bob/train} = O$

From $A \longrightarrow B$, pseudo force F = ma $\longleftarrow$ acts on bob. Let bob turns through 180° and has speed v' at highest position



W/E theorem $A \rightarrow B$:

FR = $\frac{1}{2}mv^2 + mgh$

$\frac{mv^2}{R} = 2(ma - mg)$ …….(i)

$B \longrightarrow C$: Now train has stopped, a = 0,

No pseudo force acts on bob.

Conservation of Energy

mgR + $\frac{1}{2}mv'^2 = \frac{1}{2}mv^2$

$\frac{mv'^2}{R} = \frac{mv^2}{R} - 2mg$

T + mg = 2 (ma – mg) – 2mg [From eq. (i)]

On solving $a \geq 25 m/s^2$

3K – 2 = 25 K = 9

Section – C (Maximum Marks: 12)

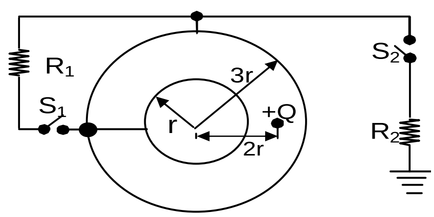
*This section contains **THREE (03)** questions stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

Consider two neutral concentric conducting thin spherical shells of radii r and 3r. A particle of positive charge Q is fixed at a distance 2r from the common center of the shells. The inner shell can be electrically connected to the outer shell through a resistance R_1 with the help of switch S_1 and outer shell can be grounded through a resistance R_2 with the help of another switch S_2 as shown.

H_1 (total heat) is dissipated in the resistance R_1 after the switch S_1 is closed keeping the switch S_2 open until a steady state is reached:



When steady state is reached after closing the switch S_1, the other switch S_2 is closed. Total Heat dissipated after the switch S_2 is closed until a new steady state is reached is H_2. (Take $\frac{Q^2}{\pi\varepsilon_0 r} = 108$ SI units]

14. Then the value of H_1 is ___________

Ans. 00000.56

Sol. $H_1 = \frac{Q^2}{192\pi\varepsilon_0 r}$

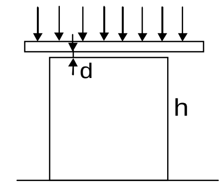
15. Then the value of H_2 is ___________

Ans. 00004.33

Sol. $H_2 = \frac{Q^2}{24\pi\varepsilon_0 r}$

Question Stem for Question Nos. 16 and 17

Question Stem



A glass plate is placed above a glass cube of 2 cm edges in such a way that there remains a thin air layer between them, see figure. Electromagnetic radiation of wavelength between 400 nm and 1150 nm (for which the plate is penetrable) incident perpendicular to the plate from above is reflected from both air surfaces and interferes. In this range only two wavelengths give maximum reinforcements, one of them is 400 nm. λ is the second wavelength. T is change in temp. necessary to warm up the cube so as it would touch the plate. The coefficient of linear thermal expansion is $\alpha = 8 \times 10^{-6}$. the refractive index of the air $\mu = 1$. The distance of the bottom of the cube from the plate does not change during warming up.
Then,

16. The value of λ (in nm) ______

Ans. 00666.67

17. The value of T (in ^oC) ______

Ans. 00003.10

Sol. $2\mu d = (2n+1)\lambda/2$ for constructive interference

$$\lambda = \frac{4\mu d}{2n+1}$$

$$\lambda = 4\mu d,\ \frac{4\mu d}{3},\ \frac{4\mu d}{5},\ \frac{4\mu d}{7} \text{-------}$$

As $n \uparrow \lambda \uparrow$

$$400 = \frac{4\mu d}{2n+1} \Rightarrow 4\mu d = 400(2n+1)$$

$$\frac{400}{2(n-1)+1} < 1150 \Rightarrow 30n > 31 \quad n > \frac{31}{30} \qquad n = 2, 3, \text{-----}$$

$$\frac{4\mu d}{2(n-2)+1} > 1150 \Rightarrow n < \frac{77}{30} \qquad n = 0,1,2 \text{ ----}$$

n = 2 (common)

$$\frac{4 \times 1 \times d}{5} = 400 \qquad d = 500 \text{ nm.}$$

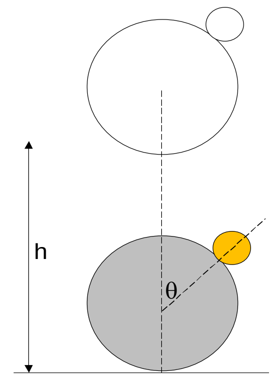
$$\Delta L = L \propto T \quad \Rightarrow T = \frac{d}{L \propto} = 3.1^oC.$$

λ = 666.67 nm

Question Stem for Question Nos. 18 and 19

Question Stem

A heavy steel ball held in contact with an extremely light steel ball at height h $=5\sqrt{3}$ m above a hard horizontal floor. The line joining the centres of the balls makes angle $\theta = 30^\circ$ with a vertical. If all the collisions are elastic and sizes of the balls are negligible as compared to the height h, the horizontal range of the lighter ball. Is R & speed Just after collision is u.
Then,



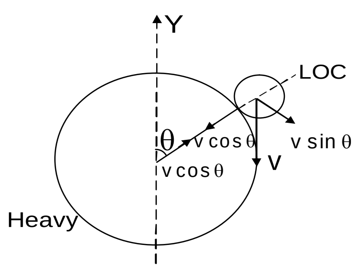
18. The value of R (in m) is ______

Ans. 00120.00

19. The value of u (in m/s) is _____

Ans. 00034.82

Sol.



Let velocity of light ball along LOC after collision is V_1

$V_1 - v\cos\theta = 2v\cos\theta$

$V_1 = 3v\cos\theta$

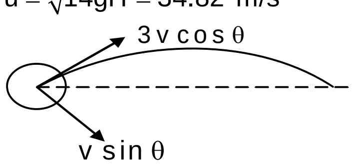
Speed u = $\sqrt{v_1^2 + (v\sin\theta)^2}$

$= \sqrt{9v^2\cos^2\theta + v^2\sin^2\theta}$

$v^2 = 2gh$

$u = \sqrt{9 \times 2gh \times \frac{3}{4} + 2 \times g \times h \times \frac{1}{4}} = \sqrt{\frac{56gh}{4}}$

$u = \sqrt{14gH} = 34.82$ m/s



Find Range R = $8h\sin 2\theta\left(4\cos^2\theta - 1\right)$

R = $8 \times 5\sqrt{3} \times \frac{\sqrt{3}}{2} \times \left(4 \times \frac{3}{4} - 1\right) = 120$m.

Chemistry

PART – II

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

20. Which of the following statement(s) is correct?

(A) Most four-coordinated complexes of Ni^{+2} ions are square planer rather than tetrahedral.

(B) The $\left[Fe\left(H_2O\right)_6\right]^{+3}$ ions is more paramagnetic than the $\left[Fe\left(CN\right)_6\right]^{-3}$ ion.

(C) Square planar complexes are more stable than octahedral complexes.

(D) The $\left[Fe\left(CN\right)_6\right]^{-4}$ ion is paramagnetic but $\left[Fe\left(CN_6\right)\right]^{-3}$ ion is diamagnetic.

Ans. ABC

Sol. $\left[Fe\left(CN\right)_6\right]^{-4}$ has Fe^{+2} and have no unpaired electron, so it is diamagnetic and $\left[Fe\left(CN\right)_6\right]^{-3}$ has Fe^{+3} and having one unpaired electron, so it is paramagnetic.

21. Regarding the structure of compound, $PBr_2Cl\left(CH_3\right)_2$, which of the following is incorrect statement(s)?

(A) Both CH_3 occupy axial position

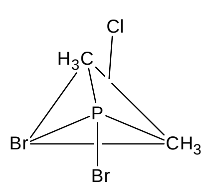
(B) Both Br occupy axial position

(C) One Cl and one Br occupy axial position.

(D) One CH_3 and one Br occupy axial position

Ans. ABD

Sol.



less electronegative atom occupy equatorial position.

22. Which of the following statement(s) is correct.

(A) In metallic carbonyls the ligand CO molecule acts both as donor and acceptor.

(B) The ligand thiosulphate, $S_2O_3^{-2}$ can give rise to linkage isomerism.

(C) The complex ferricyanide does not follow effective atomic number (EAN) rule.

(D) The complex $\left[Pt\left(Py\right)\left(NH_3\right)\left(NO_2\right)ClBrI\right]$ exist in eight geometrical forms.

Ans. ABC

Sol. In complex $\left[Pt(Py)(NH_3)(NO_2)ClBrI\right]$, 15 geometrical isomers exist.

23. In an oxide ions are placed with FCC unit cell in which B^{+3} ion occupy x% of octahedral void and A^{+2} ion occupy y% of tetrahedral void, then what is incorrect about the unit cell?

(A) Formula is $A_3B_2O_6$ in which x = 50% and y = 12.5%

(B) Formula is $A_2B_4O_6$ in which x = 50% and y = 50%

(C) Formula is AB_2O_4 in which x = 50% and y = 12.5%

(D) Formula is AB_2O_4 in which x = 12.5% and y = 50%

Ans. ABD

Sol. $A_1B_2O_4$ of x = 50% and y = 12.5%
$A_4B_2O_4$ of x = 50% and y = 50%
AB_2O_4 of x = 50% and y = 12.5%
$A_4B_{1/2}O_4$ of x = 12.5% and y = 50%

24. If $E^0_{Zn^{+2}/Zn} = -0.76V, E^0_{Ni^{+2}/Ni} = -0.24, E^0_{Fe^{+3}/Fe} = -0.04V, E^0_{Cd^{+2}/Cd} = -0.40V$
Which is/are correct statements

(A) $Zn^{+2} + Cd \longrightarrow Cd^{+2} + Zn$, spontaneous

(B) $Ni^{+2} + Cd \longrightarrow Ni + Cd^{+2}$, spontaneous

(C) $Fe^{+3} + Ni \longrightarrow Ni^{+2} + Fe$, spontaneous

(D) $Cd^{+2} + Zn \longrightarrow Zn^{+2} + Cd$, spontaneous

Ans. BCD

Sol. E^0_{cell} is (+) ve, then cell reaction is spontaneous
$E^0_{cell} = E^{(R.P)}_{cathode} - E^{o(R.P)}_{Anode}$
$= -0.76 + 0.40$
$= -0.36$ (Nonspontaneous)

25. Each of the following options contains a set of four molecules. Identify the option(s) where all four molecule possess permanent dipole moment at room temperature.

(A) $BeCl_2$, CO_2, BCl_3, $CHCl_3$

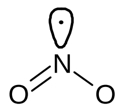
(B) NO_2, NH_3, $POCl_3$, CH_3Cl

(C) BF_3, O_3, SF_6, XeF_6

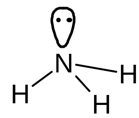
(D) SO_2, C_6H_5Cl, H_2Se, BrF_5

Ans. BD

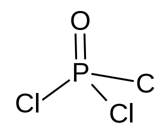
Sol.



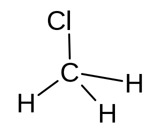
angular



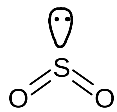
Pynomidal



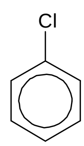
Tetrahedral but attached atom are different

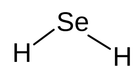


Tetrahedral but attaned atom are different

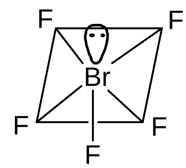


angular





Angular



square pyramidal

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 26 and 27

Imagine a universe in which the four quantum numbers can have the same possible values as in our universe except the angular momentum quantum number "ℓ" can have integral values of 0, 1, 2, 3 …….n (instead of 0, 1, 2……….(n -1) and magnetic quantum numbers "m" can have integral values of $-(\ell+1)$ to 0 to $(\ell+1)$ instead of $-\ell$ to 0 to $+\ell$.

26. Electronic configuration of the element with atomic number 25, based on the above assumption is.

(A) $1s^2, 1p^6, 2s^2, 1d^{10}, 2p^5$

(B) $1s^2, 1p^{10}, 2s^2, 1d^{11}$

(C) $1s^3, 1p^9, 2s^3, 2p^9, 3s^1$

(D) $1s^6, 1p^{10}, 2s^6, 2p^3$

Ans. D

27. Based on above assumption magnetic moment of the element with atomic number 18 is

(A) $\sqrt{24}$ B.M.

(B) $\sqrt{8}$ B.M.

(C) Zero B.M.

(D) $\sqrt{15}$ B. M.

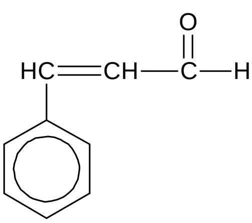
Ans. B

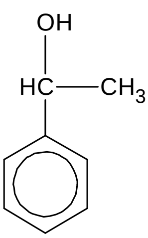
Sol. (26 – 27)
$1s^6, 1p^{10}, 2s^6, 2p^3, 2d$
$\mu_{spin-only} = \sqrt{n(n+2)} = \sqrt{2(2+2)} = \sqrt{8}$ B.M.

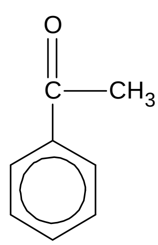
Paragraph for Question Nos. 28 and 29

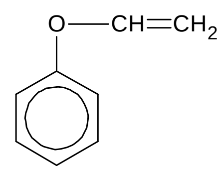
A compound (x) of molecular formula C_8H_8O is treated with two equivalent of Br_2 in ether at 0^oC in presence of a base to give a substituted product (Y) of molecular formula $C_8H_6OBr_2$. Compound (Y) on treatment with alkali followed by HCl forms α-hydroxy acid (z) of molecular formula $C_8H_8O_3$. (x) forms an oxime as well as showing positive iodoform test.

28. Compound (X) is

(A)


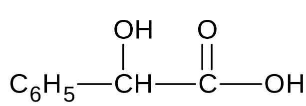
(B)


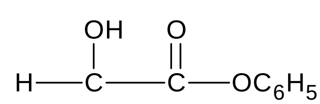
(C)


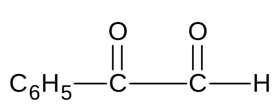
(D)


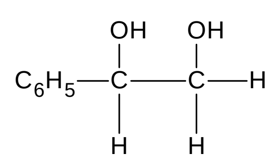
Ans. C

29. Compound (z) is

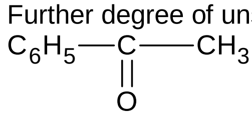
(A)


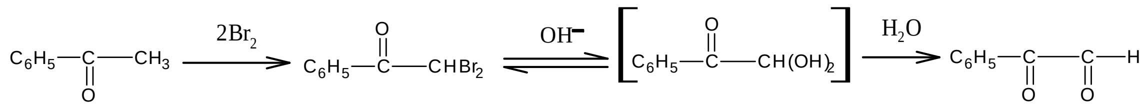
(B)


(C)


(D)


Ans. A

Sol. (for Q. 28-29)
(X) is likely to be a methyl ketone since it form an oxime as well as shown positive iodoform test. Further degree of unsaturation of (x) is 5. Therefore (X) would be




(X) (Y)

The formed compound undergoes intramolecular Cannizaro reaction.

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. In how many number of molecule/ions central element is sp^3 hybridized state.
$ClO_2^-, I_3^-, IF_6^-, BrF_6^+, ICl_2^+, CsBrCl_2, Br\left[XeO_6\right], \left(CN\right)_2$
$\left[Co\left(NH_3\right)_6\right]^{+3}, \left[Ni\left(CN\right)_4\right]^{-2}, \left[Ni\left(CO\right)_4\right]$

Ans. 3

Sol. Only in $ClO_2^-, ICl_2^+, \left[Ni\left(CO\right)_4\right]$ is sp^3 hybridized.

31. A saturated solution of a sparingly soluble salt MCl_2 has a vapour pressure of 31.78 mm of Hg at 30^oC. While pure water exert a pressure of 31.82 mm of Hg at the same temperature. Calculate the solubility product of the compound at this temp. is approximately $x \times 10^{-5} mol^3L^{-3}$. Find the value of x.

Ans. 5

Sol. Let the solubility of MCl_2 be small/lit

$$\underset{}{MCl_{2(s)}} \rightleftharpoons \underset{s}{M^{+2}_{(aq)}} + \underset{2s}{2Cl^-_{(aq)}}$$

Using Roult's law

$$\frac{P_0 - P_s}{P_0} = \frac{n}{N}$$

$$\frac{31.82 - 31.78}{31.82} = \frac{3s}{\frac{1000}{18}}$$

On solving s = 0.0232 mol/lit

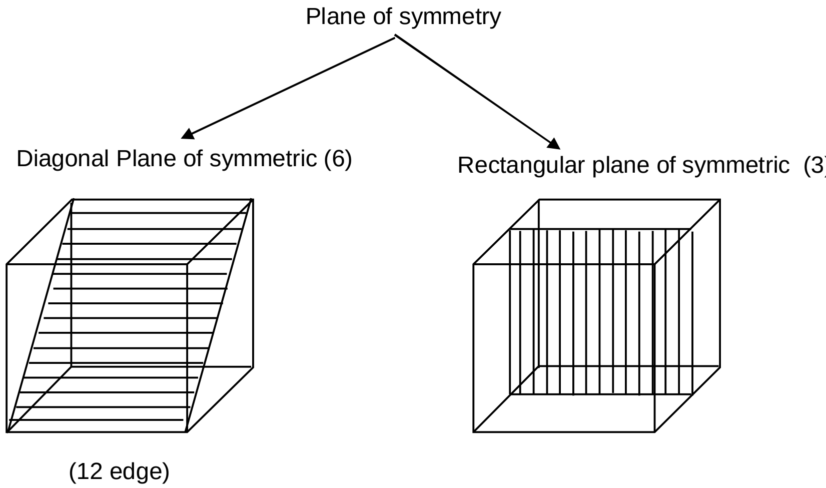
$\therefore \quad Ksp = 4s^3$

$= 4.9 \times 10^{-5} \approx 5 \times 10^{-5}$

32. What are the number of total plane of symmetry in a cube?

Ans. 9

Sol.

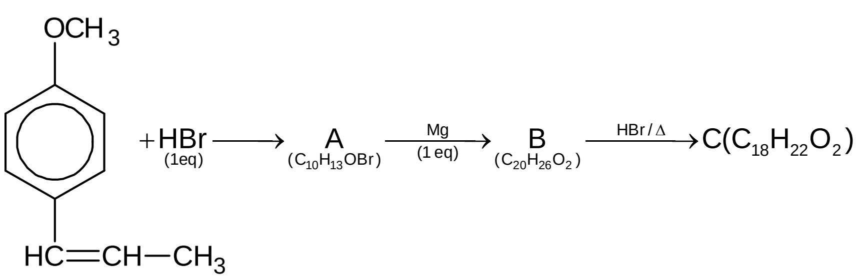


Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** questions stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem



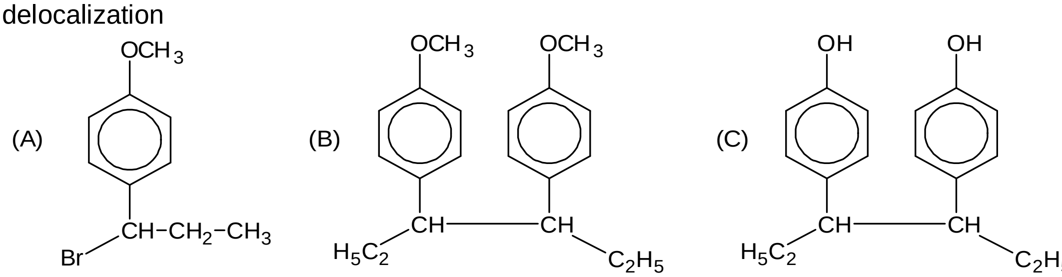
33. Degree unsaturation of compound A is ________.

Ans. 00004.00

34. Degree unsaturation of compound B is________.

Ans. 00008.00

Sol. (for Q. 33-34)
In the case H^+ from HBr gas to the double bond making it a carbocation that can be stabilized by delocalization



Question Stem for Question Nos. 35 and 36

Question Stem

Heavy metal compound (A) with a (+2) oxidation state on the central metal on heating produces a gas (B). The element of gas(B) forms a paramagnetic oxide (C). (A) on treatment with a chromate salt produces yellow crystalline p.p.t (D). (B) on reaction with water procures two acids (E), (F); (F) on heating gives (B) {Atomic weight of Pb = 207, Ba = 137}

35. Molecular weight of compound (A) is_________.

Ans. 00331.00

36. Molecular formula of (E) is HNO_x, the value of x is _______.

Ans. 00002.00

Sol. (for Q. 35-36)

$$\underset{(A)}{Pb(NO_3)_2} \longrightarrow PbO + 2\underset{(B)}{NO_2} + \frac{1}{2}O_2,\ Pb(NO_3)_2 \xrightarrow{K_2CrO_4} \underset{(D)}{PbCrO_4}$$

$$NO_2 + H_2O \xrightarrow[\text{reaction}]{\text{Disproportioration}} \underset{(F)}{HNO_3} + \underset{(E)}{HNO_2}$$

$$2\underset{(F)}{HNO_3} \longrightarrow 2NO_2 + H_2O + \frac{1}{2}O_2$$

(C) is NO

Question Stem for Question Nos. 37 and 38

Question Stem

An organic compound undergo number of parallel reaction can give product P_1, P_2, P_3 ……….P_n. The rate constant of these parallel reaction are K, 2K, 3K …….. nK and activation energy for each parallel reaction is E, 2E, 3E ……….nE respectively.

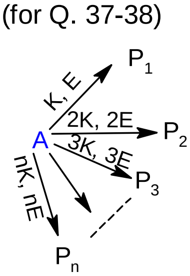
37. The overall activation energy of reaction is given by $\frac{E(2n+1)}{x}$. The value of x is_______.

Ans. 00003.00

38. A reaction follow two parallel path have activation energies 100 and 200 kJ/mol respectively. Rate constant of these parallel path are respectively $2\times10^2\ \&\ 3\times10^2\ min^{-1}$. The overall activation energy reaction in (kJ) is_____.

Ans. 00160.00

Sol. (for Q. 37-38)



Let rate cont. $K_0 = K + 2K + 3K + ...nE$

$$K_0 = Ae^{-\frac{E_0}{RT}}$$

$$\frac{dK_0}{dT} = Ae^{-\frac{E_0}{RT}}\left(\frac{E_0}{RT^2}\right)$$

$$\frac{dK_0}{dT} = K_0\frac{E_0}{RT^2}$$

$$\frac{1}{K_0}\frac{dK_0}{dT} = \frac{E_0}{RT^2}$$

$$\frac{1}{K_0}\left[\frac{d}{dT}(K + 2K + 3K + ...nK)\right] = \frac{E_0}{RT^2}$$

$$\frac{1}{K_0}\left[\frac{K.E}{RT^2} + \frac{2K.2K}{RT^2} + \frac{3K.3E}{RT^2} + ...\frac{nk.nE}{RT^2}\right] = \frac{E_0}{RT^2}$$

$$\frac{K.E}{RT^2}\left[1^2 + 2^2 + 3^2 + ...n^2\right] = \frac{E_0}{RT^2}$$

$$\frac{K.E}{K(1+2+3+.....n)}\frac{(n)(n+1)(2n+1)}{6} = E_0$$

$$\frac{n(n+1)}{\frac{n(n+1)}{2}}\frac{(2n+1)}{6} = E_0$$

$$\boxed{E_0 = \frac{(2n+1)}{3}E}$$

Overall activation anergy of two parallel path is given by

$$E = \frac{K_1E_1 + K_2E_2}{K_1 + K_2}$$

$$E = \frac{\left(100 \times 2 \times 10^2 + 200 \times 3 \times 10^2\right)}{2 \times 10^2 + 3 \times 10^2}$$

$$= \frac{800 \times 10^2}{5 \times 10^2}$$

$$= 160$$

Mathematics PART – III

Section – A (Maximum Marks: 24)

This section contains ***SIX (06)*** *questions. Each question has* ***FOUR*** *options (A), (B), (C) and (D).* ***ONE OR MOER THAN ONE*** *of these four option(s) is (are) correct answer(s).*

39. If D_1, D_2, D_3,……..D_{1000} are 1000 doors and $P_1, P_2 P_3$............P_{1000} are 1000 persons. Initially all doors are closed. Changing the status of doors means closing the door if it is open or opening it if it is closed. P_1 changes the status of all doors. Then P_2 change s the status of D_2, D_4, D_6,........ D_{1000} (Doors having number which are multiples of 2). Then P_3 changes the status of D_3, D_6, D_9 . ...D_{999} (doors having number which are multiples of 3) and this process is continued till P_{1000} changes the status to D_{1000}, then the doors which are finally open is/are

(A) D_{961}

(B) D_{269}

(C) D_{413}

(D) D_{729}

Ans. AD

Sol. D_m will be open finally if m is the perfect square

40. $f(x) = \cos\pi(|x| +[x])$, then

(A) f is continuous at x = 1/2

(B) f is continuous at x = 0

(C) f is differentiable in (-1, 0)

(D) f is differentiable in (0, 1) .

Ans. ACD

Sol. $f(x) = \cos\pi(|x| +[x])$
$f(0) =1$, $f(0 – 0) = \cos\pi(-1) = –1$
$f(1/2) = \cos\pi\left(\frac{1}{2}\right) = 0$, $f\left(\frac{1}{2}-0\right) = \cos\pi\left(\frac{1}{2}\right) = 0$,
$f\left(\frac{1}{2}+0\right) = \cos\frac{\pi}{2} = 0$
for $x \in (0, 1)$ $f(x) = \cos\pi x$
for $x \in (-1, 0)$ $f(x) = \cos\pi (-x- 1) = - \cos\pi x$.

41. If z_1, z_2, z_3, z_4 are complex numbers in an Argand plane satisfying $z_1 + z_3 = z_2 + z_4$. A complex number 'z' lies on the line joining z_1 and z_4 such that $\text{Arg}\left(\frac{z-z_2}{z_1-z_2}\right) = \text{Arg}\left(\frac{z_3-z_2}{z-z_2}\right)$. It is given that $|z-z_4| = 5, |z-z_2| = |z-z_3| = 6$ then

(A) area of the triangle formed by $z, z_1 z_2$ is $3\sqrt{7}$ sq. units

(B) area of triangle formed by z, z_3, z_4 is $\frac{15\sqrt{7}}{4}$ sq. units

(C) area of the quadrilateral formed by the points z_1, z_2, z_3, z_4 taken is order is $\frac{27\sqrt{7}}{2}$ sq. units

(D) area of the quadrilateral formed by the points z_1, z_2, z_3, z_4 taken is order is $\frac{47\sqrt{7}}{3}$ sq. units

Ans. ABC

42. Which of the following is/are correct?

(A) If A is a $n \times n$ matrix such that $a_{ij} = \left(i^2 + j^2 - 5ij\right) \cdot (j - i) \forall$ I and j then trace $(A) = 0$

(B) If A is a $n \times n$ matrix such that $a_{ij} = \left(i^2 + j^2 - 5ij\right) \cdot (j - i) \forall$ I and j then trace $(A) \neq 0$

(C) If P is a 3×3 orthogonal matrix, α, β, γ are the angle made by a straight line with OX, OY, OZ and $A = \begin{bmatrix} \sin^2\alpha & \sin\alpha \cdot \sin\beta & \sin\alpha \cdot \sin\gamma \\ \sin\alpha \cdot \sin\beta & \sin^2\beta & \sin\beta \cdot \sin\gamma \\ \sin\alpha \cdot \sin\gamma & \sin\beta \cdot \sin\gamma & \sin^2\gamma \end{bmatrix}$ and $Q = P^T AP$, then $PQ^6 P^T = 32A$

(D) Id matrix $A = \left[a_{ij}\right]_{3\times3}$ and matrix $B = \left[b_{ij}\right]_{3\times3}$ where $a_{ij} + a_{ji} = 0$ and $b_{ij} - b_{ji} = 0 \ \forall$ i and j then $A^6 B^7$ is a singular matrix

Ans. ACD

Sol. (A) $a_{ij} = -a_{ji} \Rightarrow a_{ij} + a_{ji} = 0 \Rightarrow$ A is skew symmetric matrix
(B) $A^2 = 2A \Rightarrow A^3 = 2^2 A \Rightarrow A^6 = 2^5 A$
(D) $A^6 B^7$ is a skew symmetric matrix of odd order

43. If three normals can be drawn from the point (h, 0) to the parabola $y^2 = -4x$, then h can be

(A) any real number in the interval $(-\infty, -3)$

(B) any real number in the interval $(-4, -3)$

(C) any real number in the interval $(-\infty, -4)$

(D) any real number

Ans. ABC

Sol. (h, 0) satisfy the equation of the normal $y = mx - 2am - am^3$

44. The solution of $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$ satisfying y(1) = 1 is given by :

(A) a system of hyperbola

(B) a system of circles

(C) $y^2 = x(1 + x) - 1$

(D) $(x - 2)^2 + (y - 3)^2 = 5$

Ans. AC

Sol. Put $y^2 = u$

We get $\frac{du}{dx} - \frac{u}{x} = \frac{1}{x} + x$, which is in linear form.

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 45 and 46

Let y = f(x) is a curve and P(x, y) be any point on it. The tangent and the normal drawn at P meets x-axis at T and N respectively PT and PN are called length of tangent and length of normal at P. If M is the foot of perpendicular drawn from P on x-axis, then MT and MN are called length of sub-tangent and length of sub-normal at P. Answer the following questions.

45. The equation of the curves passing through (0, 1), if the length of sub-normal at any point of the curve remains unity is

(A) $y^2 = \pm x + 1$

(B) $y^2 = \pm \frac{x}{2} + 1$

(C) $y^2 = \pm 2x + 1$

(D) $y^2 = \pm 3x + 1$

Ans. C

Sol. $\left| y\frac{dy}{dx} \right| = 1 \Rightarrow y^2 = \pm 2x + 1$

46. The equation of the curve passing through (1, 2), if the intercepts of normal on x-axis is equal to thrice of the abscissa of point P is

(A) $3x^2 - y^2 + 1 = 0$

(B) $x^2 - 2y^2 + 7 = 0$

(C) $2x^2 - y^2 + 2 = 0$

(D) $x^2 - y^2 + 3 = 0$

Ans. C

Sol. $x + y\dfrac{dy}{dx} = 3x$; solving we get $2x^2 - y^2 + 2 = 0$.

Paragraph for Question Nos. 47 and 48

If $m > 0$, $n > 0$, the definite integral $I = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$ depends upon the values of m and n and is denoted by β (m, n), called the beta function.

e.g. $\int_0^1 x^4(1-x)^5\,dx = \int_0^1 x^{5-1}(1-x)^{6-1}\,dx = \beta(5,\ 6)$ and

$\int_0^1 x^{5/2}(1-x)^{-1/2}\,dx = \int_0^1 x^{7/2-1}(1-x)^{1/2-1}\,dx = \beta\left(\frac{7}{2},\ \frac{1}{2}\right)$. Obviously, β (n, m) = β (m, n).

47. The integral $\int_0^{\pi/2} \cos^{2m}\theta \sin^{2n}\theta\, d\theta$ is equal to

(A) $\frac{1}{2}\beta\left(m + \frac{1}{2},\ n + \frac{1}{2}\right)$

(B) 2β (2m, 2n)

(C) β (2m + 1, 2n + 1)

(D) None of these

Ans. A

Sol. Writing $\sin^2\theta = x$, we get $2\sin\theta\cos\theta\, d\theta = dx$, and hence the given integral is equal to

$$\frac{1}{2}\int_0^{\pi/2} \cos^{2m-1}\sin^{2n-1}\theta\,(2\sin\theta\cos\theta)\,d\theta$$

$$= \frac{1}{2}\int_0^1 \left(\cos^2\theta\right)^{\frac{2m-1}{2}}\left(\sin^2\theta\right)^{\frac{2n-1}{2}}dx = \frac{1}{2}\int_0^1 (1-x)^{m-1/2}x^{n-1/2}dx = \frac{1}{2}\beta\left(m + \frac{1}{2},\ n + \frac{1}{2}\right).$$

48. If $\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx = k\int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx$, then k is equal to

(A) $\frac{m}{n}$

(B) 1

(C) $\frac{n}{m}$

(D) None of these

Ans. B

Sol. Writing $\frac{x}{1+x} = z$, we get $x = \frac{z}{1-z}$, $1 + x = \frac{1}{1-z}$ and $dx = \frac{dz}{(1-z)^2}$.

L.H.S. = $\int_0^1 \frac{z^{m-1}}{(1-z)^{m-1}} (1-z)^{m+n} \frac{dz}{(1-z)^2} = \int_0^1 z^{m-1}(1-z)^{n-1}\,dz = \beta\,(m, n)$

$= \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx$.

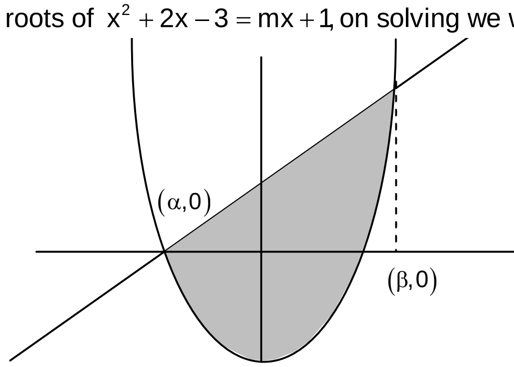
Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. If the least value of the area bounded by the line y = mx +1 and the parabola $y = x^2 + 2x - 3$ is α where m is a parameter then the value of $\frac{6\alpha}{32}$ is ________

Ans. 2

Sol. $A = \int_\alpha^\beta (y_1 - y_2)\,dx$ where α, β are the roots of $x^2 + 2x - 3 = mx + 1$, on solving we will get



$\frac{1}{6}\left(m^2 - 4m + 20\right)^{3/2}$: Hence $\alpha = \frac{32}{3}$

$\Rightarrow \frac{6\alpha}{32} = 2$

50. The shortest distance between $(1-x)^2 + (x-y)^2 + (y-z)^2 + z^2 = \frac{1}{4}$ and 4x+2y+4z+7=0 in 3 - dimensional coordinates system is equal to _________

Ans. 2

Sol. Let $a = 1 - x$

$b = x - y$

$c = y - z$

$d = z$

then a+b+c+d=1 and $a^2+b^2+c^2+d^2=\frac{1}{4}$

$\Rightarrow (a-b)^2+(a-c)^2+(a-d)^2+(b-c)^2+(b-d)^2+(c-d)^2=0$

$\Rightarrow a=b=c=d$

$\therefore x=\frac{3}{4}, y=\frac{1}{2}, z=\frac{1}{4}$

So the distance from the point $\left(\frac{3}{4},\frac{1}{2},\frac{1}{4}\right)$

From the plane $4x+2y+4z+7=0$ is $\frac{3+1+1+7}{6}=2$

51. The equation of the plane passing through the intersection of the planes $2x-5y+z=3$ and $x+y+4z=5$ and parallel to the plane $x+3y+6z=1$ is x+3y+6z=k, where k is

Ans. 7

Sol. Equation of plane passing through the intersection of the planes $2x-5y+z=3$ and x+y+4z=5 is

$(2x-5y+z-3)+\lambda(x+y+4z-5)=0$

$\Rightarrow (2+\lambda)x+(-5+\lambda)y+(1+4\lambda)-3-5\lambda=0$ ....(i)

Which is parallel to the plane $x+3y+6z=1$.

Then $\frac{2+\lambda}{1}=\frac{-5+\lambda}{3}=\frac{1+4\lambda}{6}$

Then, $\frac{2+\lambda}{1}=\frac{-5+\lambda}{3}=\frac{1+4\lambda}{6}$

$\therefore \lambda=\frac{-11}{2}$

from eq. (i)

$-\frac{7}{2}\times-\frac{21}{2}y-21z+\frac{49}{2}=0$

$\therefore x+3y+6z=7$

Hence, k = 7

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** questions stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Experiment -1: Let 'S' be the sample space obtained when two different coloured unbiased coins are tossed. Two subsets P and Q (of two elements each) are randomly chosen from S one after another with replacement.

Element obtained in P and Q are recorded.

Experiment -2: The same two coins used in the previous experiment are tossed. The outcomes are recorded.

If the outcomes obtained in experiment 2 matches with nay of the outcomes of experiment 1, then both the experiments are declared 'successful'.

(Note that experiment '2' is conducted after experiment 1)

52. Probability that 'success' is from 'P' but not from 'Q'

Ans. 00000.25

Sol. $\text{Case}-1: n(P \cap Q) = 0$

Let S_P denote 'success' from P, then

$$P(S_P \cap \bar{S}_Q) = \frac{1}{2} \times P\left(\frac{E_1}{n(P \cap Q) = 0}\right) = \frac{1}{2} \times \frac{{}^4C_2}{\left({}^4C_2\right)^2}.$$

$\text{Case}-2: n(P \cap Q) = 1 \Rightarrow n(P \cup Q) = 3$

$$P(S_P \cap \bar{S}_Q) = \frac{1}{4} \times P\left(\frac{E_1}{n(P \cap Q) = 1}\right) = \frac{1}{4} \times \frac{{}^4C_3 \times {}^3C_1 \times 2}{\left({}^4C_2\right)^2}.$$

$\text{Case}-3: n(P \cap Q) = 2$

$$P(S_P \cap \bar{S}_Q) = 0$$

$$P(S_P \cap \bar{S}_Q) = \frac{2}{12} + \frac{1}{12} = \frac{1}{4}.$$

53. Probability that there is no 'success' if experiment 1 is conducted after experiment 2

Ans. 00000.25

Sol. $\text{Case}-2: n(P \cap Q) = 0$

$\Rightarrow n(P \cup Q) = 4$

$\Rightarrow P(\bar{S}_P \cap \bar{S}_Q) = 0$

$\text{Case}-2: n(P \cap Q) = 1 \Rightarrow n(P \cup Q) = 3$

$$P(\bar{S}_P \cap \bar{S}_Q) = \frac{{}^3C_1 \times 2}{\left({}^4C_2\right)^2} = \frac{1}{6}.$$

$\text{Case}-3: n(P \cap Q) = 2 \Rightarrow n(P \cup Q) = 2$

$$P(\bar{S}_P \cap \bar{S}_Q) = \frac{{}^3C_2}{\left({}^4C_2\right)^2} = \frac{1}{12}.$$

$$P(\bar{S}_P \cap \bar{S}_Q) = \frac{2}{12} + \frac{1}{12} = \frac{1}{4}.$$

Question Stem for Question Nos. 54 and 55

Question Stem

Let $\vec{r}$ is a position vector of a variable point in Cartesian OXY plane such that $\vec{r} \cdot (10\hat{j} - 8\hat{i} - \hat{r}) = 40$ and $p_1 = \max\left\{\left|\vec{r} + 2\hat{i} - 3\hat{j}\right|^2\right\}, p_2 = \min\left\{\left|\vec{r} + 2\hat{i} - 3\hat{j}\right|^2\right\}$. A tangent line is drawn to the curve $y = \frac{8}{x^2}$ at the point A with abscissa 2. The drawn line cuts x-axis at a point B.

54. p_2 is equal to

Ans. 00003.40

Sol. $p_2 = \min\left|\vec{r} + 2i - 3j\right|^2$

where $\vec{r}\cdot\left(10\hat{j} - 8\hat{i} - \hat{r}\right) = 40$

take $\vec{r} = x\hat{i} + y\hat{j}$

$\left(x\hat{i} + y\hat{j}\right)\cdot\left(\left(10 - y\right)\hat{j} - \left(8 + x\right)\hat{i}\right)$

$\Rightarrow x\left(8 + x\right) - y\left(10 - y\right) = -40$

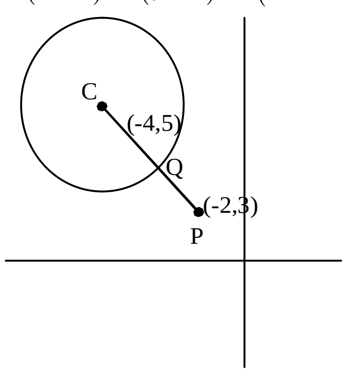
$x^2 + y^2 + 8x - 10y + 40 = 0$

$\left(x + 4\right)^2 + \left(y - 5\right)^2 = 1$ (i)

$\left|x\hat{i} + y\hat{j} + 2\hat{i} - 3\hat{j}\right|^2$

$\Rightarrow \left|\left(x + 2\right)\hat{i} + \left(y - 3\right)\hat{j}\right|^2$

$= \left(x + 2\right)^2 + \left(y - 3\right)^2 = \left(\text{minimum distance of } \left(x, y\right) \text{from } \left(-2, 3\right)\right)^2$



$= \left(PQ\right)^2$

$= \left(PC - 1\right)^2$

$= \left(2\sqrt{2} - 1\right)^2$

$= 8 + 1 - 4\sqrt{2} = 9 - 4\sqrt{2}$

55. $\vec{AB}\cdot\vec{OB}$ is

Ans. 00003.00

Question Stem for Question Nos. 56 and 57

Question Stem

Let $\Delta \neq 0$ and Δ^C denotes the determinant of cofactors, then $\Delta^C = \Delta^{n-1}$, where n (>0) is the order of Δ.

56. If a, b, c are the roots of the equation $x^3 - px^2 + r = 0$, then the value of
$\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$ is (if p = 0.9)

Ans. 00000.53

Sol. $\therefore a+b+c = p, ab+bc+ca = 0, \Rightarrow a^2+b^2+c^2 = (a+b+c)^2 - 2(ab+bc+ca) = p^2$

If $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ then $\Delta^C = \begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix} = \Delta^{3-1} = \Delta^2$

$$\Rightarrow \Delta^C = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$= \begin{vmatrix} a^2+b^2+c^2 & ab+bc+ca & ab+bc+ca \\ ab+bc+ca & a^2+b^2+c^2 & ab+bc+ca \\ ab+bc+ca & ab+bc+ca & a^2+b^2+c^2 \end{vmatrix} = \begin{vmatrix} p^2 & 0 & 0 \\ 0 & p^2 & 0 \\ 0 & 0 & p^2 \end{vmatrix}$$

57. If a, b, c are the roots of the equation $x^3 - 3x^2 + 3x + 7 = 0$, then the value of
$\begin{vmatrix} 2bc-a^2 & c^2 & b^2 \\ c^2 & 2ac-b^2 & a^2 \\ b^2 & a^2 & 2ab-c^2 \end{vmatrix}$ is

Ans. 00000.00

Sol. $(x-1)^3 = -8 \Rightarrow x-1 = -2, -2\omega, -2\omega^2$

$\Rightarrow x = -1, 1-2\omega, 1-2\omega^2$

$\therefore a = -1, b = 1-2\omega, c = 1-2\omega^2$

$$\because \begin{vmatrix} a & b & c^2 \\ b & c & c \\ c & a & b \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}\begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix} = \begin{vmatrix} 2bc-a^2 & c^2 & b^2 \\ c^2 & 2ac-b^2 & a^2 \\ b^2 & a^2 & 2ab-c^2 \end{vmatrix}$$